Shivanandham Sankaranarayanan

Senior Software Engineer · AI Engineering (RAG, Agents, MCP) & Backend Architecture

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SUMMARY

Senior Software Engineer with 5+ years building production backend, data and AI systems for fintech, SaaS and media clients. Led the design and build of an AI-native engineering platform (Claude Code agents, MCP integrations, golden-flow evals) adopted by a ~20-engineer team, a production Retrieval-Augmented Generation (RAG) assistant on FastAPI, pgvector and LangChain, and an AWS Personalize recommendation engine with automated retraining. Deep in Python (FastAPI, Django), Ruby on Rails, PostgreSQL, Celery and Kafka; owns systems from schema design to CI/CD and production support.

SKILLS

AI / LLM systems	RAG pipelines, embeddings, pgvector, semantic search, reranking, citation grounding, hallucination controls, LangChain, LangGraph, Pydantic AI, structured outputs, prompt design, OpenAI / Azure OpenAI, Anthropic Claude, Google Gemini, Mistral
Agentic development	Claude Code (agents, skills, hooks, workflows), MCP servers (author and operator), multi-agent orchestration, LLM evals (golden flows, artifact-based judging), tracing and telemetry hooks, Cursor
ML & data pipelines	AWS Personalize, MLOps (automated retraining, incremental refresh), Redshift, S3, Kinesis Firehose, Lambda, ETL, web scraping at scale (Playwright, rate-limited crawlers), entity resolution (pg_trgm)
Backend	Python (FastAPI, Django + DRF, SQLAlchemy, Celery), Ruby on Rails, Node.js (Express), Kafka producers, REST API design, JWT auth and RBAC, webhooks, background jobs
Data stores & infra	PostgreSQL (pgvector, pg_trgm), Redis, AWS Timestream, Elasticsearch, AWS, GCP, Azure, Cloudflare Workers, Docker, GitHub Actions, Sentry, Datadog
Frontend & mobile	Vue 3, Nuxt 3, React, Next.js, TypeScript, Flutter, Tailwind CSS

EXPERIENCE

Francium Tech — Senior Software Engineer

Jan 2021 – Present

Consulting engineer embedded in client teams; every engagement below was delivered as a Francium Tech engineer.

MoneySmart (fintech, SG/HK) — Lead engineer: AI tooling, recommendations, platform — Nov 2022 – Present

Claude Code, MCP, AWS Personalize, Redshift, S3, Kinesis Firehose, Lambda, Kafka, Ruby on Rails, Nuxt 3, Vue 3 + TypeScript, Cloudflare Workers, GrowthBook

- **Agent Smith (Feb 2026 – Present):** led the design and build of MoneySmart's AI-native engineering control plane, where engineers drive Claude Code agents from a one-line feature request to reviewed, tested pull requests across 50+ service repositories; adopted by a ~20-engineer team.
- Designed the multi-agent workflow (research, planning, implementation, review and QA agents with explicit contracts under a tier-0 orchestrator; 11 agents, 10 workflows, 16 skills) and a 19-domain knowledge layer spanning Rails APIs, Nuxt frontends and Kafka event flows.
- Built the evaluation harness: golden-flow tests judged on artifacts (gate ledgers, session traces, gh call logs) rather than assistant prose, plus a triage eval over 40 real engineer prompts; shipped PR gating via a guard hook in shadow mode before enforcing it.
- Wired governed MCP access to 12 production surfaces (Redash, Sentry, Datadog, CircleCI, Cloudflare, Playwright) with trace/telemetry hooks and permission guardrails.
- **Recommendation engine (Nov 2022 – 2026):** led the AWS Personalize recommendation system for SG and HK: Redshift extraction to S3, dataset imports, Kinesis Firehose event capture, Lambda-backed serving, automated daily incremental refreshes and scheduled full retrains (MLOps).
- Layered business logic on the model (viewed-product boosts, post-purchase category de-boosts, cold-start lists, popularity scoring, life-stage and socio-demographic segments); built the V4 affinity engine with caching and channel redistribution and fixed its CPU hot spot; A/B-tested output via GrowthBook.
- **Platform services:** Kafka producers across the user-data, membership, campaign and interstitial services (account closure, consent, email capture, claim approval, earnings history) and the Braze sync backfill; campaigns, rewards and membership Rails APIs and admin tools; Nuxt + Strapi landing-page builder; shared Nuxt core/auth modules.
- Current focus, product platform: Rails product-info and product-listing APIs (bulk key-features grid, product families), Nuxt 3 comparison pages with JSON-LD/SEO, a Vue 3 + TypeScript product admin replacing the legacy tool, GrowthBook A/B routing on Cloudflare Workers.

Golden Set Analytics (tennis scouting) — Data and predictive-analytics platform — Apr 2026 – Aug 2026

Python, Django 5 + DRF, Celery + Redis, PostgreSQL (pg_trgm), Playwright, Docker, Azure

- Built the data-acquisition layer across 6 public sources (CoreTennis, ITF, WTN, ATP/WTA, ITA, Sackmann): a token-bucket rate limiter shared by 8 parallel workers at 10–20 req/s, retry/backoff honouring Retry-After, and self-healing past Cloudflare and Incapsula blocks with headless Playwright.

- Designed the two-layer schema (raw per-source intake promoted into a source-neutral core) with an orientation-independent match identity for cross-source de-duplication, and pg_trgm player-identity resolution with confidence bands that auto-routes ~95% of records and sends ambiguous merges to review.
- Cut a per-row promotion check from ~123 ms to under 1 ms at 800K rows with partial indexes and shrank the matcher candidate pool from 500 to 100 as the core passed 1M players; contributed to the Elo rating and player-trajectory feature layer behind the prediction model.

Teldat (networking vendor, Spain) — Tech lead, AI product assistant (RAG) Oct 2025 – Mar 2026

FastAPI, PostgreSQL + pgvector, SQLAlchemy/Alembic, LangChain, LangGraph, Celery + Redis, OpenAI, Mistral, GCS, React 19, GitHub Actions, Sentry

- Tech lead for a six-engineer team on a RAG assistant that answers product and solution questions from Teldat's documentation and generates PPT, Excel and DOCX sales documents; owned architecture, delivery and production operations.
- Built ingestion for 7 source types (sitemaps, PDF/DOCX/PPTX manuals, external URLs and files, SharePoint, YouTube transcripts, bulk Excel) orchestrated with Celery, with embeddings on pgvector and citation tracking on every answer.
- Tuned retrieval relevance, added hallucination controls (grounded link suggestions, no invented URLs) and manual knowledge injection, and built authentication and RBAC with JWT refresh-token rotation and fine-grained permissions.
- Owned production readiness: GitHub Actions deploys with release listing and one-click rollback, Sentry, Celery/Flower tuning and database connection-pool leak fixes; device-fingerprint visitor tracking and admin analytics; 429-test suite.

Mainspring (creative agency) — AI ad-content generation platform Jul 2025 – Nov 2025

FastAPI, SQLAlchemy/Alembic, Celery + Redis, Google Gemini via Pydantic AI, Stability AI, Airtable, React 19 + shadcn

- Top backend contributor on a platform that syncs agency campaign plans from Airtable and generates ad copy (Gemini with structured outputs via Pydantic AI, three options per frame) and banner imagery (Stability AI outpainting).
- Built retry-safe Celery generation tasks with backoff and per-component status, the outpaint geometry solver, and the external API that feeds assets into Figma, After Effects and Photoshop; added service-level tests that helped raise backend coverage from 60% to 75%.

Undivided (parent-support platform, US) — RAG Q&A assistant Mar 2025 – Nov 2025

FastAPI, PostgreSQL + pgvector, OpenAI / Azure OpenAI, Next.js

- Built a RAG question-answering assistant for parents of children with disabilities: retrieval and re-ranking over pgvector, feedback loops on answers, and SEO-focused, accessible Next.js pages.

B-Cube.ai (crypto trading platform) — Led the v2 rebuild Sep 2024 – Mar 2025

Ruby on Rails 8, Vue 3 + Vite, PostgreSQL, AWS Lambda, AWS Timestream, Sumsub, Chargebee, Etherscan

- Led the rebuild of a crypto trading platform from Node.js/Angular to a Rails 8 API and Vue 3 frontend, including migration of legacy FusionAuth users, 2FA and KYC-gated authorisation.
- Built signal fan-out into per-user executions placing orders via AWS Lambda across 8 exchanges (Binance real-time WebSocket, KuCoin, Kraken, Bybit and more; spot and futures), plus bot and portfolio analytics as parameterised SQL over Timestream (returns, drawdown, profit factor).
- Integrated Sumsub KYC and Chargebee subscriptions with idempotent webhooks, and BCUBE ERC-20 token payments verified on-chain via Etherscan.

Earlier engagements — Teldat (internetworking), Hybr1d (HR SaaS), Soulsearch (matchmaking) Jan 2021 – Aug 2024

React frontends and Ruby on Rails APIs for three production products; RSpec-tested; direct client communication.

PROJECTS

- **multi-database-mcp** (Oct 2025; Node.js, MCP SDK; published on npm, MIT) — github.com/shivanandham/multi-database-mcp. Model Context Protocol (MCP) server giving AI coding agents 9 tools to list, switch, inspect and query multiple PostgreSQL databases per project; used daily across client work.
- **BixOps restaurant POS** (Dec 2025 – Present; Rails 8.1, Next.js, Flutter) — tiered POS for small food businesses: offline Flutter billing apps with Bluetooth printing and a multi-tenant cloud platform with kitchen order tickets, a real-time kitchen display (ActionCable), gapless GST-compliant invoicing via row locks and a plan/entitlements engine; 248-spec backend suite.
- **Free LLM gateway** (Jun 2026; Python) — OpenAI-compatible gateway that routes across 10 free-tier LLM providers (Groq, Cerebras, Gemini, OpenRouter, Mistral, SambaNova and more), picks the best available model and fails over with cooldowns on rate limits and quota exhaustion.
- **Luma pregnancy companion** (Oct – Nov 2025; Flutter, Node.js/Express, Prisma, Gemini 2.5 Flash) — tracking app with AI daily checklists and a chat that triages first, auto-logs symptoms and retrieves remembered facts into later prompts.
- **DreamersKit AR video memories** (Oct 2026; FastAPI, OpenCV, React, MindAR + three.js) — printed photos play their linked video when scanned in the browser: SIFT + RANSAC recognition finds the photo, WebAR tracks it and plays the video on it; an admin app prints QR sheets.
- **Biometric attendance terminal** (Mar – Apr 2025; ESP32, C++/PlatformIO, Rails 8) — fingerprint device with server-driven enrolment and rollback and per-company device keys, wired to a multi-tenant employee-management API.

EDUCATION

B.Tech, Computer Science — Hindustan Institute of Technology and Science, Chennai 2017 – 2021

(capstone: powered wheelchair prototype)